CHAPTER 5
Demand for Labour in Competitive Labour Markets

Answers to End-of-Chapter Questions:

1. (LO1,2) This is an interior solution where the isoquant is tangent to the isocost line. The marginal rate of technical substitution between labour and capital is equal to the ratio of their prices. Any other point in the diagram is either unattainable for the firm, or is sub-optimal, meaning that the firm can make higher profits by a substitution of one input for another. The firm's expansion path is the locus of points (capital/labour combinations) that the firm chooses as it expands production. Do not confuse it with the labour demand curve, which is in employment-wage space. The expansion path is always in labour-capital space, which is just like the isocost and the isoquant curves. Normally, we would expect the expansion path to be positively sloped - as output expands, the firm hires more labour. It need not be a straight line, however, and in general it is not.

2. (LO3) For an inferior factor of production, as the level of output increases (decreases), the quantity demanded of the factor falls (increases), all other factors held constant. How would the firm's demand for labour be altered if labour were an inferior factor of production? The scale effect of a wage decrease would be negative (production increases, so the quantity demanded of labour decreases for an inferior factor), but the substitution effect would be positive, working in the direction of an increase in the quantity demanded of labour. The quantity demanded of labour would still increase. The labour demand curve still has a negative slope. Note than in this case, the two effects work in opposing directions.

3. (LO3) This statement is false. See the end of the section entitled 'the demand for labour in the short run'. The theory of labour demand assumes that labour is homogenous in quality. The negative relationship is due to a negative scale effect and a negative substitution of a wage change on the quantity demanded of labour.

4. (LO3, LO4) The labour demand curve for the monopolist is steeper than it is in the case for the firm which is a competitor in the output market, and generally the demand will be less elastic. The demand for labour is equal to the marginal revenue product of labour. If the output market is competitive, this marginal revenue product is called the value of the marginal product, and is equal to the product of the price and the marginal product. If the output market is uncompetitive, this marginal revenue product is equal to the product of the marginal revenue and the marginal product. Since marginal revenue is less than price at any given level of output, the wage paid to labour at any given level of labour input must be lower than what would be the case in a competitive labour market. This implies that the demand curve for labour for the monopolist is steeper.

5. (LO4) Unit labour costs are obtained by dividing labour cost by the output level. In the context of international competitiveness, there are three components to unit labour costs, which are indicated by the top line in Figure 5.8. First, there is the exchange rate. A strong Canadian dollar in the early 1990s raised the unit labour costs relative to those of some of its trading partners, and a depreciating dollar thereafter had the opposite effect. Since about 2002, the Canadian dollar has appreciated a great deal, which has driven up unit labour costs considerably. The next factor is the labour costs per hour expressed in Canadian dollars, where are indicated by the solid line in Figure 5.8. In the late 1990s they fell relative to those in our trading partners, which was due primarily to the depreciating Canadian dollar. The third component is the level of physical productivity expressed in terms of output per hour of labour time. Since 1980 Canada has lost a lot of ground relative to the United States as far as this variable is concerned. That factor has caused a major erosion in competitiveness of Canadian producers relative to their US-based counterparts.

   The figures of Table 5.1, which show the levels of compensation costs relative to Canadian levels, indicate that Canada is reasonably competitive in terms of compensation costs relative to other industrialised countries. In addition, the growth in hourly compensation (after converting into US dollars) between 1990-2009 was in line with most of Canada's trading partners (Table 5.2). In the first column of Table 5.2, it is apparent that Canada fared badly compared to most of its trading partners in terms of productivity, as productivity grew slowly between 1990 and 2009. Combining the three factors of the exchange rate, the growth in productivity, and the hourly compensation, Canada's per unit labour costs grew in the middle range of these countries over this period (the higher the decline, the greater the gain in competitiveness). Canada did not lose much competitiveness in relation to its largest trading partner, the United States. Although the performance on the productivity variable was disappointing, the fairly slow rise in hourly compensation combined with a depreciating Canadian dollar worked to prevent an erosion in competitiveness.

6. (LO4) High value-added means that the value received for the output is much higher than the cost of the inputs, so that a high level of resources accrues to the producer. In this situation, the producer earns a high margin, earning a return for the know-how, the sophisticated machinery, and the risk embodied in the product or service, which often is quite innovative (there is a definite advantage to being there first) and customised. Most economists would agree with this statement. The key to maintaining living standards is to promote productivity growth, increase production in areas in which Canada has a comparative advantage, and decrease production in certain areas in which it operates at a comparative disadvantage. More analysis related to this question appears in the response to the next question.

7. (LO4) This statement might have been true in the late 1980s and early 1990s, but due in part to a weak Canadian dollar, Canadian goods and services were quite competitive with American products in the late 1990s and the early 2000 years. By 1996, however, that

competitive advantage was gone. That was manifested in a series of very strong trade surpluses. One cannot sustain, however, a gain in competitiveness based on a continual depreciation of the currency, and that can serve to divert attention away from the continual challenge of increasing productivity, which can be achieved by modernising production facilities by investment in capital and restructuring, etc. Indeed, since 2002 the Canadian dollar has strengthened considerably, which has caused an erosion in competitiveness. The major challenge facing Canadian producers is to raise productivity. The last section of the chapter talks about the constant presence of all demand side factors, and not just international trade, in necessitating cost efficiency in production - a flexible, highly trained, and adaptable labour force, flexible, modular factories, just-in-time inventory and delivery system, etc.

8. (LO4) John Kerry was definitely assuming that less skilled labour located in the US is a substitute for labour of a similar grade in those countries where the production is outsourced. To the extent that that act of substitution is taxed, the US-based labour becomes more competitive with the foreign-based labour, thus giving firms less of an incentive to make that choice of out-sourcing production. Recall, however, that there should always be a scale effect that works in favour of US-based labour. As production is outsourced, the costs of production decline, which should work (as least to some degree) to expand output of these firms. As long as low-skilled labour in the US is a normal rather than an inferior factor of production, that factor in isolation will raise the demand for both foreign labour and domestic labour. Therefore, outsourcing will not hurt domestic labour quite as much as it would appear on the surface. If domestic labour is actually a complement to foreign labour, which is not impossible under all circumstances, then Kerry's proposal will backfire.

Answers to End-of-Chapter Problems:

1. (LO1)
   a) Given that the production function is $Q = 2L^{1/2}$, we can derive the function for the marginal product of labour, which is the derivative of that function with respect to labour input: $1/L^{1/2}$. The next step is to derive the function for the marginal revenue product of labour. Assuming that the product market is perfectly competitive, the MRP = VMP = Price*marginal product of labour, which works out to: $10/L^{1/2}$ (given a wage of \$10.00). Since the firm's demand curve for labour is the same as its MRP curve (provided that we are below the firm's average revenue product curve), it follows that the expression for the demand curve is $W = 10/L^{1/2}$. Solving for L, we have $L = 100/W^2$. Note that there is a negative relationship between L and W, as there should be.

   b) The values for the second column can be obtained by simply inputting the suggested value for the wage that appears in the first column. Once one has obtained these values for labour input, they can be inputted into the production function $Q = 2L^{1/2}$ to obtain the values for the third column. In order to obtain the values for profit, it is necessary to ignore the costs incurred by the other factors of production. These figures are thus interpreted as the profits net of labour costs. The expression is: total revenue - total labour costs = Price*output - wage*labour input = 10*output (from column 3) - wage (from column 1)*labour input (from column 2). The figures appear in the table below.

Wage	Labour demand	Output	Profits
$1.00	100	20	$100
$2.00	25	10	$50
$5.00	4	4	$20
$10.00	1	2	$10

The profit level falls as the wage increases. At each of these points, the firm is maximizing profits given the wage.

2. (LO1, 2, 3) This type of production technology is called Leontief, or the fixed-coefficients, production technology. One day of labour is combined with one day of hammer use to produce one unit of output, and no substitution among inputs is possible. A hammer is useless without a corresponding labourer, and vice versa. The isoquants are L-shaped in capital-labour space. The vertex of this right angle is at the point (100,100), where capital goes on the vertical axis and labour goes on the horizontal axis. The interpretation of the isoquant is that any increase in labour input without a corresponding increase in capital input produces no additional helmets, and vice versa. The only way to reach higher levels of output, and thus higher level isoquants, is to increase both inputs in the one-to-one proportions that are given. If the firm must produce 100 units of output, it needs 100 days of labour and 100 days of hammer use. The labour costs are $5,000 and the hammer costs are $1,000, which makes a total cost of $6,000. If the wage changes to $25 per day, then the labour costs fall by half to $2,500, and the capital costs remain at $1,000. There is no change in the use of these factors, however. The firm would like to substitute capital for labour, as the price of labour has fallen by 50%, but it cannot, given the rigidity of the technology of production. The substitution effect is thus zero. There might be a scale effect, however. For the same outlay of $6,000, for firm could produce approximately 170 helmets, and thus hire 170 units of labour with 170 units of capital. That increase in the quantity demanded of labour would be attributable to a scale effect.

3. (LO3) The answer to this question depends on whether the output level is fixed or not. If the case mentioned in the textbook under the heading “Labour Demand Under Cost Minimization” applies, the level of output is not generated by profit maximizing behaviour. The employer is likely to minimise costs subject to a certain level of service being provided. Unless the services of doctors and the services of nurses are perfect

complements, which means that they are not substitutable at all, the statement is true. Place the level of nursing services on one axis of the isoquant-isocost diagram, and the level of doctor services on the other axis. As the wage paid to nurses increases, the relative wage paid to doctors decreases, and the isocost line will rotate along a given isoquant. The health care organisation will re-allocate its labour input toward doctors and away from nurses. This is due to a pure substitution effect.

In the case in which the level of output is variable, and the health care provider sets this level by profit maximization behaviour, an increase in the wage paid to nurses will shift the marginal cost of production curve upward. This will normally result in a lower level of output (a lower level of provision of health acre services). Due to the scale effect, the quantity demanded for doctors or well as for nurses will fall. On the other hand, the relative wage of doctors has fallen, so due to the substitution effect, the quantity demanded for doctors will increase. The net effect of this wage change on the quantity demanded for doctors depends on the magnitudes of the scale effect and the substitution effect, which work in opposite directions. If the scale effect dominates, the statement is false, but if the substitution scale effect dominates, the statement is true.

4. (LO1, 2, 3)

a) Unlike the case in problem #2, for which there was zero substitutability between the two factors of production, there is perfect substitutability between these two inputs of child labour and adult labour. The isoquants are linear functions. If L_A appears on the vertical axis and L_C appears on the horizontal axis, the slope of each isoquant is -1/ 2, reflecting the fact that adults are twice as efficient as children. For example, for the isoquant corresponding to 10 units of output, the intercepts are (1, 0) and (0, 1/2).

b) The isocost lines are linear functions. If L_A appears on the vertical axis and L_C appears on the horizontal axis, the slope of each isocost line is - 3/5, reflecting the fact that adult labour is 2/3 more expensive than child labour (or that child labour costs only 3/5 as much as adult labour). Intuitively, one can tell at this stage that the employer will hire only adults, because adults are twice as productive, but do not cost twice as much as children (if the price of child labour was $50, then the firm would be indifferent between choosing between the two of them).

c) The equation for the isoquant is $100 = 2\ L_A + L_C$. If the price of child labour is $60, 50 adults are hired, and no children are hired. The total production cost is $5,000. Any other combination of L_A and L_C would cost more. The production point occurs at (0, 50) where the isocost line and the isoquant intersect at the vertical axis. If the price of child labour falls to $25, then all isocost lines have an equation of the form D (a constant) = $100L_A + 25L_C$, and the slope is -1/4. There is no change in the isoquants. In this case, the profit maximizing input occurs at the point (100, 0), at the point of intersection of the isoquant, the isocost line, and the horizontal axis. No adults and 100 children are hired, and the total cost of producing 1,000 units is $2,500. In this case, there is a substitution effect of infinity, as firms respond to the decrease in the price of child labour by replacing all of the adult labour with child labour. There would also be a scale effect if the firm decided to increase output. If the firm wanted to spend the original value of $5,000, it could produce 2,000 bushels of apples, for which it would hire 200 children. The demand curve for child labour has a kink in it. For any wage that is greater than 50% of the adult wage, the unit labour cost (per bushel of apples) for children is higher than it is for adults, and thus no children are hired. The demand curve coincides with the vertical axis for all wages above $50. For all wages below $50, the unit labour cost is lower for children, but the exact amount of labour hired is indeterminate (depending on how many apples it could sell in the output market). You can thus draw the labour demand curve as infinitely elastic at that wage.

5. (LO4) The information that is provided regarding a wage of $15.00 per hour is irrelevant to this question.

   a) Chemical herbicides can be thought of as a substitute factor for the task of weeding, which requires manual labour. If they are banned from a jurisdiction, the firm does not have a viable substitute for the factor of manual labour, which works to reduce the wage elasticity of demand.

   b) If the product market in which the coal is sold is characterized by a monopoly, the wage elasticity of demand is likely to be inelastic, ceteris paribus.

   c) This case is similar to the previous question. The price elasticity of demand in the output market for cigarettes is lower than it is in the output market for fast food, so ceteris paribus, we would expect for the wage elasticity of demand to be higher (more elastic) in the latter.

   d) Given the technology of production for firms in Silicon Valley, they are likely to employ many more computer programmers than secretaries. This implies that the payroll for the former comprises a much higher share of total production costs than does the payroll for the latter. Ceteris paribus, this implies that the wage elasticity of demand for programmers is likely to be more elastic than it is for secretaries. Note that we expect computer programmers to be paid more than secretaries because the programmers have a higher value of marginal revenue product. That is not the question here, however. For this question, we want to know whether the quantity demanded for programmers is more responsiveness to wage changes than is the case for secretaries. The answer should be yes. While secretaries are paid less, the wage elasticity of demand for their services is lower.

   e) It is probably harder to replace quality control engineers, either with machinery or with other workers, than it is to replace beer production line workers. Ceteris paribus, this implies that the wage elasticity of demand for the tasters is likely to be more elastic than it is for the beer production line workers. Nevertheless, a lot of you would probably like a job that involved tasting beer. I am assuming, for this question, that that job involves a lot of training.

6. (LO4)

a) For Canada, the opportunity cost of producing one more unit of food is 2 autos foregone, while for USA the opportunity cost of producing one more unit of food is 1 auto foregone. For Canada, the opportunity cost of producing one more auto is 1/ 2 unit of food foregone, while for USA the opportunity cost of producing one more auto is 1 unit of food foregone. Since the opportunity cost of more food production is lower in the USA, and the opportunity cost of more auto production is lower in Canada, Canada has a comparative advantage in the production of autos, while the United States has a comparative advantage in the production of food. Canada should specialise completely in the production of autos, and trade them for its food. The United States should specialise completely in the production of food, and trade it them for its autos. This trade pattern will allow both countries to consume at points which are beyond their production possibility frontiers, or the transformation curves. Wages will increase in terms of the units of food and units of autos that can be consumed by workers – i.e., their living standards, or their consumption possibilities. All of the workers in Canada's food industry lose their jobs, but in theory many if not most of them should be able to find work in the expanding auto sector.

b) Pay particular attention to the references to the work by Gaston and Trefler. Most of the economics profession believes that the FTA and NAFTA have been beneficial in the aggregate for Canada's economy. Currently, exports comprise at least 40% of Canada's GDP, and almost 90% of them are purchased by the United States.

c) This statement is totally false, as it pays no attention to all of the other economic forces that were influencing employment in Canada over that period. Ceteris paribus, we expect that trade liberalization will increase employment. Almost all economists believe that that was the case.

7. (LO3) Refer to Figure 5.6.

For the case of substitutes: If two factors of production are substitutes, then one can be used in place of the other. As the wage rises, the quantity demanded for labour will fall as long as the labour demand curve is not perfectly inelastic (i.e., vertical). We will move up and along a given labour demand curve. If labour and capital are substitutes in the production process, the quantity demanded for capital will rise. In the longer term, as the capital stock increases from K_1 to K_0, the capital to labour ratio rises, and labour becomes more productive at the margin. Assuming a perfectly competitive output market, the short run labour demand curve shifts to the right. At any given wage, the quantity demanded of labour rises. Some or all of the initial decrease in employment that arose from the initial wage increase (which was large) is reversed.

For the case of complements: If two factors of production are complements, then they are used together. One cannot use more or less of one without using more (or less) of the other. As the wage increases, there is a negative scale effect that causes a small decrease in the quantity demanded of labour (there is no substitution effect). If labour and capital are complements in the production process, the quantity demanded for capital will fall along with the quantity demanded of labour. In the longer term, as the capital stock decreases from K_0 to K_1, the capital to labour ratio falls, and labour becomes less productive at the margin. Assuming a perfectly competitive output market, the short run labour demand curve shifts to the left. At any given wage, the quantity demanded of labour falls. The initial decrease in employment that arose from the initial wage increase (which was small) is amplified in this case.

Note that the long run labour demand curve (MRP_N (LR)) does not shift at all. That curve takes into account different values for the capital stock. It is the two short-run labour demand curves that shift, as they are drawn holding the stock of capital fixed. This question deals with a somewhat tricky point. If capital and labour are complements in the production process, the degree of substitutability between them is low. This works to make the short-run wage elasticity of labour demand inelastic, as measured along any short-run labour demand curve, all other factors held constant. In the long run, however, changes in the capital stock will cause the short run labour demand curve to shift, and what becomes relevant is the long-run wage elasticity of labour demand, which is measured along the long-run labour demand curve. This quantity is more elastic than is the case along the short-run demand curves.